

Synergizing Federated Learning and Differential Privacy: A Comprehensive Survey on Architectures, Threats, Mechanisms, and Trustworthiness

1 Introduction

The proliferation of distributed machine learning has catalyzed a paradigm shift from centralized data repositories to decentralized training architectures, yet this transition alone fails to guarantee data confidentiality. While Federated Learning (FL) mitigates the risks of raw data exposure by exchanging only model updates, it remains vulnerable to sophisticated inference attacks that can reconstruct sensitive training samples from shared gradients. As noted in [1], the assumption that decentralization inherently ensures privacy is a dangerous misconception; adversaries can exploit shared parameters to launch gradient inversion, membership inference, and property inference attacks. Consequently, the integration of Differential Privacy (DP) has emerged as a critical necessity, providing rigorous, mathematically provable guarantees against such leakage. However, reconciling the competing objectives of privacy, utility, and communication efficiency presents a formidable trilemma. The injection of calibrated noise, essential for satisfying (ϵ, δ) -DP definitions, inevitably degrades model convergence and accuracy, a trade-off that is particularly acute in deep learning contexts [2].

This survey critically examines the synergistic potential of combining FL with DP to address these escalating privacy risks. We move beyond the superficial layer of parameter exchange to analyze the fundamental tension between formal privacy guarantees and model utility. Recent empirical evaluations suggest that many existing DP implementations utilize privacy budgets (ϵ) so large that they offer meaningless theoretical protection while severely hampering utility [3]. Conversely, stringent privacy settings often render models unusable, highlighting the “no free lunch” theorem where privacy preservation necessitates a quantifiable sacrifice in utility or efficiency [4; 5]. Our analysis dissects this trilemma, exploring how advanced composition theorems, such as Rényi Differential Privacy and the moments accountant, allow for tighter privacy accounting over multiple communication rounds compared to traditional methods [6; 7].

Furthermore, we delineate the scope of this survey to encompass not merely algorithmic mechanisms but the entire ecosystem of trustworthy private learning. We categorize architectural variants, distinguishing between horizontal, vertical, and transfer learning settings, each presenting unique privacy exposure surfaces and requiring tailored DP mechanisms [8]. The discussion extends to the interplay between DP and other system objectives, including robustness against poisoning attacks and fairness in heterogeneous data distributions. While DP noise can inadvertently amplify biases against underrepresented groups, it also offers certified robustness against certain adversarial manipulations [9; 10].

The organizational structure of this work is designed to guide the reader from foundational definitions to emerging frontiers. We begin by rigorously defining threat models and privacy granularities, distinguishing between user-level and record-level protection [11]. Subsequent sections synthesize algorithmic innovations, including adaptive noise scheduling and privacy amplification via shuffling and subsampling [12]. We further investigate hybrid architectures that integrate cryptographic primitives like Secure Multi-Party Computation and Trusted Execution Environments with DP to create multi-layered defenses [13]. Finally, we address domain-specific applications in healthcare and finance, alongside regulatory compliance challenges under frameworks like GDPR and the EU AI Act [14]. By bridging theoretical bounds with empirical auditing methodologies, this

survey aims to provide a comprehensive roadmap for developing next-generation privacy-preserving systems that are both mathematically sound and practically viable.

2 Foundational Architectures, Threat Models, and Privacy Definitions

2.1 Architectural Paradigms and Data Partitioning Strategies in Distributed Learning

The architectural landscape of federated learning (FL) is fundamentally defined by the topology of data partitioning, which dictates the communication protocols, threat surfaces, and the specific instantiation of differential privacy (DP) mechanisms required to ensure confidentiality. While the overarching paradigm shifts computation to the data source to mitigate the risks of centralization, the assumption that decentralization alone guarantees privacy is demonstrably false; shared model updates remain vulnerable to sophisticated inference and reconstruction attacks [15]. Consequently, the choice between horizontal, vertical, and transfer learning architectures necessitates distinct privacy-preserving strategies tailored to the underlying data distribution and feature alignment.

Horizontal Federated Learning (HFL) addresses scenarios where clients share a common feature space but possess disjoint sample sets, a setting prevalent in cross-device applications involving mobile users or IoT sensors. In HFL, the primary privacy objective is often client-level DP, which protects the presence of an entire user’s dataset within the training cohort rather than individual records [16]. However, the aggregation of gradients in HFL does not inherently prevent membership inference or property leakage, as adversaries can exploit the statistical signatures embedded in shared parameters [9]. The efficacy of DP in HFL is further complicated by statistical heterogeneity (Non-IID data), where uniform noise injection can disproportionately degrade model utility for clients with sparse or skewed distributions, necessitating adaptive clipping and personalized privacy budgeting [17].

In contrast, Vertical Federated Learning (VFL) arises when parties hold complementary feature sets for the same set of samples, a common arrangement in cross-silo collaborations between financial institutions and healthcare providers [18]. VFL introduces unique vulnerabilities distinct from HFL, particularly regarding intermediate representation leakage. Since parties must exchange embeddings or gradients to align features, adversaries can launch feature inference or label inference attacks by analyzing these intermediate values, even without access to raw data [19; 20]. Therefore, privacy mechanisms in VFL must extend beyond gradient perturbation to include secure multi-party computation protocols that mask intermediate activations, alongside feature-level DP to sanitize the latent representations exchanged during forward and backward passes [21; 22].

When both sample sets and feature spaces are disjoint, Federated Transfer Learning (FTL) becomes necessary to bridge domain gaps through knowledge transfer. FTL complicates privacy preservation by requiring the alignment of disparate distributions, often necessitating the exchange of prototype vectors or domain-invariant representations that can inadvertently leak source domain characteristics [23]. Recent advancements in foundation models have revitalized FTL via parametric feature transfer, where clients share synthetic features or adapter parameters rather than raw gradients, offering a new avenue for integrating DP with reduced sensitivity [24]. Furthermore, emerging hybrid architectures, such as SplitFed, amalgamate FL with split learning to optimize the trade-off between communication efficiency and privacy exposure, though they require rigorous auditing to prevent reconstruction attacks on the split boundaries [25].

Ultimately, the selection of an architectural paradigm fundamentally shapes the adversary’s capability and the requisite defense mechanism. While HFL focuses on obscuring individual contribu-

tions within a swarm, VFL and FTL must secure the semantic integrity of shared representations against active inference. Future architectural designs must therefore move beyond static partitioning assumptions, incorporating dynamic, privacy-aware topology adaptation that accounts for the evolving threat landscape in heterogeneous distributed environments [26].

2.2 Formal Privacy Guarantees, Adversarial Capabilities, and Trust Models

Building upon the architectural constraints and trust boundaries delineated in the preceding section, this subsection establishes the rigorous mathematical frameworks underpinning privacy in distributed learning, specifically contrasting the trust assumptions and adversary capabilities inherent in Local, Central, and Shuffled Differential Privacy (DP) models. While the decentralization of data storage in Federated Learning (FL) mitigates centralization risks, it does not inherently guarantee confidentiality against the sophisticated inference attacks previously highlighted [27]. Consequently, formal DP guarantees are essential to bound the information leakage from shared model updates. The taxonomy of DP models in FL is primarily defined by the trust placed in the aggregator. Central DP (CDP) assumes a trusted server that aggregates raw updates before adding noise, offering strong utility but failing catastrophically if the server is compromised [28]. In contrast, Local DP (LDP) mandates that clients perturb their updates locally prior to transmission, ensuring privacy even against a malicious aggregator, though often at the cost of significant utility degradation due to the accumulation of noise [29]. Recent advancements have introduced the Shuffled Model, which leverages secure shuffling protocols to break the link between users and their updates, thereby amplifying privacy guarantees to approach CDP levels while maintaining the trust assumptions of LDP [30].

The efficacy of these models is contingent upon a precise definition of adversarial capabilities, which directly informs the specific attack vectors detailed in the subsequent section. Adversaries in FL are categorized based on their position and intent: honest-but-curious servers that follow the protocol but attempt to reconstruct private data from gradients; malicious clients that deviate from the protocol to poison the global model or infer properties of other participants; and external eavesdroppers monitoring communication channels [31]. The threat landscape is further complicated by collusion, where multiple parties (e.g., a subset of clients and the server) coordinate to breach privacy. Research indicates that standard secure aggregation protocols may remain vulnerable to reconstruction attacks if the server controls a sufficient number of Sybil devices to isolate a target user’s contribution [32]. Furthermore, active adversaries can exploit the iterative nature of training to launch gradient inversion or membership inference attacks, recovering raw inputs or determining dataset inclusion with high confidence [33; 34].

Addressing these threats requires robust mechanisms for collusion resistance and Sybil resilience. The integration of Secure Multi-Party Computation (SMPC) with DP offers a pathway to verify update integrity without exposing individual contributions, though this introduces complex trade-offs between communication overhead and privacy amplification [35]. Notably, privacy amplification via subsampling and shuffling provides a theoretical mechanism to tighten privacy bounds (ϵ) without increasing noise magnitude, provided that client participation is random and uncorrelated [36]. However, empirical auditing reveals a gap between theoretical worst-case bounds and practical leakage, suggesting that adaptive adversaries may exploit implementation flaws or non-IID data distributions to exceed predicted privacy losses [3; 37]. Future research must therefore focus on dynamic trust models that adapt to varying adversary capabilities and develop verification protocols capable of detecting collusion in real-time, ensuring that formal guarantees hold under realistic, heterogeneous deployment scenarios [38]. This transition from theoretical bounds to em-

pirical validation sets the stage for the following analysis of specific attack vectors that exploit the high-dimensional geometry of shared gradients.

2.3 Advanced Inference, Reconstruction, and Property Leakage Attacks

The assumption that parameter exchange in federated learning inherently preserves data confidentiality has been decisively refuted by a proliferation of sophisticated inference and reconstruction attacks. These adversarial strategies exploit the high-dimensional geometry of shared gradients to reverse-engineer sensitive training inputs, challenging the foundational trust models of distributed optimization. At the forefront of this threat landscape are gradient inversion attacks, which formulate data reconstruction as an optimization problem to recover raw inputs from shared updates. Seminal work demonstrated that deep models under collaborative learning are susceptible to Generative Adversarial Network (GAN)-based attacks that generate prototypical samples indistinguishable from the true training distribution [33]. Recent advancements have escalated this threat from approximate recovery to exact reconstruction; for instance, active adversaries can deploy “trap weights” to manipulate global models, enabling perfect data recovery from large mini-batches with negligible computational overhead [39]. Furthermore, the “Cocktail Party Attack” leverages Independent Component Analysis to separate individual inputs from aggregated gradients, effectively breaking privacy even in large-batch settings previously thought secure [40]. Theoretical analyses confirm that under mild conditions, training data is fully identifiable from a single gradient query, rendering naive parameter sharing fundamentally unsafe [41].

Beyond pixel-perfect reconstruction, inference attacks target the statistical presence and attributes of data records. Membership inference attacks (MIA) determine whether a specific sample contributed to the training set, exploiting the overfitting tendencies of stochastic gradient descent. In the white-box setting, adversaries analyze activation outputs and gradient dynamics to trace training records with high precision, even in well-generalized models [34]. This risk extends to the user level, where malicious servers can distinguish specific client identities and recover user-specified private data through multi-task discriminators, posing a severe threat to individual privacy in mobile edge computing [42]. In vertical federated learning (VFL) contexts, where parties share disjoint features, the threat surface expands to include label and feature inference. Adversaries can deduce sensitive labels from forward intermediate embeddings or reconstruct missing features using only prediction outputs, bypassing protections that focus solely on gradient perturbation [19; 20]. The CAFE attack further demonstrates that catastrophic data leakage is possible in VFL even with large batch sizes, recovering high-quality batch data from aggregated gradients [43].

Property inference attacks represent a more subtle yet pervasive risk, aiming to extract global properties of the training distribution unrelated to the main task, such as demographic attributes or dataset provenance. Research indicates that collaborative updates inadvertently leak such side information, allowing adversaries to infer when specific individuals appear in training photos or to determine the distributional characteristics of a client’s local data [15]. These attacks remain potent even against defensive measures like gradient compression or quantization, which often provide a false sense of security by failing to obscure semantic information [44]. Moreover, the efficacy of these attacks is not limited to static analysis; active adaptive strategies allow adversaries to modify model parameters in real-time to maximize leakage, as seen in “Decepticons” attacks on language models that exploit transformer architectures to retrieve high-fidelity text [45]. While differential privacy offers theoretical bounds, empirical audits reveal that standard implementations often require impractical noise levels to thwart these advanced attacks, highlighting a critical gap between theoretical guarantees and practical robustness [3]. Consequently, the field must move

beyond mere parameter obscuration toward rigorous, provable defenses that account for active, adaptive adversaries capable of exploiting the nuanced information leakage inherent in deep learning optimization dynamics.

2.4 Empirical Privacy Auditing, Lower Bounds, and Evaluation Frameworks

Building on the critical gap between theoretical guarantees and practical robustness identified in the analysis of advanced attacks, this subsection examines empirical privacy auditing as a vital methodology to validate Differential Privacy (DP) mechanisms in Federated Learning (FL). While DP provides rigorous upper bounds on information leakage, its practical efficacy often diverges from worst-case analyses due to implementation nuances and conservative accounting. Consequently, empirical auditing has emerged to bridge the divide between formal definitions and realized security. A primary approach involves adversary instantiation, where hypothetical attackers are simulated to establish lower bounds on privacy loss. [46] demonstrates that by constructing specific distinguishers for DP-SGD, one can derive tight lower bounds that match theoretical upper bounds, revealing that standard analyses are often optimal unless additional assumptions about adversary capabilities are introduced. However, [3] highlights a significant disparity in complex learning tasks, noting that settings offering strong theoretical guarantees often yield useless models, whereas practical utility frequently relies on privacy parameters that offer meaningful but theoretically weak guarantees.

To address the computational inefficiency of traditional auditing, which often requires thousands of retraining runs, recent innovations propose “one-shot” estimation techniques. [47] introduces a framework capable of systematically estimating privacy loss during a single training run, eliminating the need for extensive retraining while maintaining provable correctness under the Gaussian mechanism. Complementing this, the CANIFE framework utilizes carefully crafted “canary” samples to measure empirical exposure per communication round. [37] shows that in realistic FL scenarios, the empirical ϵ derived from canary insertion can be 4-5 times lower than theoretical bounds, suggesting that worst-case analyses may significantly overestimate practical leakage. Furthermore, information-theoretic approaches offer an alternative quantification metric. [48] employs mutual information (MI) to derive upper bounds on dataset leakage through aggregated updates, theoretically proving that privacy leakage reduces linearly with the number of participating users, a finding validated empirically via MI neural estimators.

Standardized evaluation frameworks are essential for reconciling the privacy-utility-communication trilemma inherent in these systems. [5] formulates this trade-off through a unified information-theoretic lens, establishing a “No-Free-Lunch” theorem that quantifies the lower bounds for privacy leakage, utility loss, and efficiency reduction across various protection mechanisms. Similarly, [9] provides a comprehensive measurement methodology to empirically assess how Local and Central DP variants defend against backdoor and membership inference attacks, revealing that while DP mitigates white-box membership inference, it may fail against property inference. Despite these advances, challenges remain in validating defenses against the sophisticated reconstruction attacks detailed previously. [49] utilizes generative adversarial priors to demonstrate that private data can still be leaked under certain defense settings, underscoring the necessity of stress-testing systems against adaptive adversaries. Future research must therefore focus on developing dynamic auditing tools that can operate in heterogeneous, cross-silo environments, ensuring that theoretical guarantees translate into robust, deployable privacy protections.

3 Algorithmic Mechanisms for Differentially Private Federated Optimization

3.1 Fundamental Perturbation Mechanisms and Sensitivity Control

The integration of differential privacy (DP) into federated optimization fundamentally relies on two synergistic operations: bounding the sensitivity of individual contributions and injecting calibrated stochastic noise. In the federated context, where clients transmit model updates rather than raw data, the primary challenge is to ensure that the aggregation of these updates satisfies formal privacy guarantees against inference attacks. The cornerstone of this process is gradient norm clipping, which strictly limits the influence of any single client’s data on the global model update. Without such bounding, the sensitivity of the aggregation function remains unbounded, rendering the addition of finite noise insufficient to guarantee privacy. As demonstrated in [16], clipping local updates is a prerequisite for achieving client-level differential privacy, effectively hiding individual contributions during the training phase. However, the selection of the clipping threshold presents a critical hyperparameter dilemma; a threshold set too low introduces significant bias by truncating informative gradients, while a threshold set too high necessitates excessive noise to maintain the privacy budget, thereby degrading model utility.

To address the sensitivity bounding challenge, recent advancements have moved beyond static clipping norms toward adaptive mechanisms. The work [50] proposes estimating the clipping threshold online at a specified quantile of the update norm distribution, utilizing a negligible portion of the privacy budget to dynamically track the optimal bound. This approach mitigates the need for exhaustive hyperparameter tuning and adapts to the heterogeneous nature of client data distributions. Furthermore, [51] extends this concept to coordinate-wise adaptive clipping, theoretically demonstrating that such granularity allows for the addition of less noise compared to global clipping methods, thus preserving higher model accuracy. These adaptive strategies are particularly vital in non-IID settings, where the distribution of update norms varies significantly across clients and training rounds, as noted in [52], which highlights the complex relationship between clipping bias and the convergence of clipped FedAvg algorithms.

Once sensitivity is bounded, the second pillar involves the strategic injection of calibrated noise, typically drawn from Gaussian or Laplacian distributions. The choice of mechanism dictates the privacy-utility trade-off. While the Gaussian mechanism is standard for (ϵ, δ) -DP, alternative discrete mechanisms have emerged to accommodate communication constraints and secure aggregation protocols. For instance, [30] introduces a discretized approach that matches the accuracy of central DP with low-precision communication, while [53] offers a discrete alternative based on Poisson variables that is closed under summation, making it highly efficient for secure aggregation environments. The magnitude of this noise is inversely proportional to the sensitivity bound and directly proportional to the privacy budget consumption. Consequently, the interplay between clipping and noise injection defines the feasible region of the privacy-utility frontier.

Emerging trends suggest a shift toward amplitude-varying perturbation strategies that adjust noise levels dynamically throughout the training lifecycle. [54] argues that persistent noise amplitude is suboptimal, proposing geometric series-based noise scheduling to prevent premature convergence. Additionally, the integration of privacy mechanisms with optimization landscapes is gaining traction; [55] illustrates that combining clipping with sharpness-aware minimization can yield flatter loss landscapes, enhancing robustness against the perturbations inherent to DP. Despite these advances, the fundamental tension remains: rigorous privacy guarantees often come at the cost of convergence speed and final accuracy, necessitating continued innovation in adaptive sensitivity control and noise calibration to make DPFL viable for large-scale deployment.

3.2 Privacy Amplification via Stochastic Protocols and Shuffling

Building upon the foundational mechanisms of clipping and noise injection described previously, privacy amplification via stochastic protocols represents a paradigm shift in differentially private federated learning (DPFL). While the prior section established that rigorous privacy guarantees often incur substantial utility costs under strict local trust assumptions, stochastic protocols leverage inherent randomness in client participation and data ordering to tighten privacy bounds beyond what is achievable through noise injection alone. Traditional Local Differential Privacy (LDP) typically imposes prohibitive utility costs due to the requirement of adding substantial noise at the user level to protect against an untrusted server. However, theoretical advancements demonstrate that when local randomizers are composed within a larger protocol involving subsampling or shuffling, the resulting privacy guarantee is amplified, effectively bridging the gap between local and central differential privacy models.

The mechanism of amplification by subsampling relies on the uncertainty regarding whether any specific individual’s data is included in a given computation batch. In federated settings, this is realized through random client selection strategies. [56] establishes that increasing the number of participating clients while maintaining a fixed privacy level can improve convergence, yet optimal scheduling strategies are required to balance this with privacy budgets. More significantly, [57] introduces a distributed protocol where clients make independent, randomized participation decisions without server coordination. This “random check-in” approach achieves privacy-accuracy trade-offs comparable to centralized subsampling, crucially removing the need for the server to know the total population size or initiate communication, thereby enhancing robustness in dynamic networks. Furthermore, [58] integrates these concepts into variance-reduced algorithms like DP-SCAFFOLD, demonstrating that advanced accounting under data heterogeneity allows for tighter bounds than standard DP-FedAvg when local updates are frequent.

Complementing subsampling, the shuffled model of differential privacy introduces an anonymous intermediary that permutes client updates before aggregation, breaking the link between specific updates and their originators. [59] proposes efficient protocols where communication and error scale polylogarithmically with the number of users, utilizing an “invisibility cloak” technique to make individual data indistinguishable from noise while preserving the exact sum. This approach is particularly potent in high-dimensional settings; [60] addresses the challenge of large model parameters by proposing gradient sparsification (SS-Topk) combined with shuffling, achieving accuracy improvements of over 60% compared to pure LDP baselines. The efficacy of shuffling extends to personalized settings as well, where [61] quantifies heterogeneous contributions to central privacy, showing that shuffling can amplify privacy even when clients enforce varying local privacy levels.

Despite these advantages, practical deployment faces challenges regarding trust assumptions and adversarial capabilities. While shuffling reduces the noise required, it typically assumes an honest-but-curious shuffler; if the shuffler colludes with the server, guarantees degrade to the local model. Moreover, [62] warns that malicious servers might exploit protocol inconsistencies to bypass secure aggregation, suggesting that cryptographic guarantees must be tightly coupled with differential privacy mechanisms. Recent work on [30] highlights the complex tension between communication constraints, discrete noise addition, and privacy amplification, noting that secure aggregation itself can provide a form of privacy amplification akin to shuffling if the number of users is sufficiently large.

Future directions must address the integration of these stochastic protocols with robustness against poisoning attacks, as amplified privacy can sometimes mask malicious updates. Additionally, ex-

tending amplification theories to non-IID data distributions and adaptive participation rates remains critical. By synergizing stochastic participation, secure shuffling, and advanced composition theorems, the field moves toward protocols that offer central-level utility with only local-level trust assumptions, fundamentally altering the privacy-utility frontier in distributed optimization. However, the iterative nature of these stochastic protocols necessitates rigorous tracking of the cumulative privacy loss over multiple communication rounds to ensure these amplification effects are accurately quantified, a challenge that motivates the detailed exploration of refined accounting frameworks in the following section.

3.3 Advanced Compositional Accounting and Cumulative Privacy Tracking

In differentially private federated learning (DPFL), the iterative nature of stochastic gradient descent necessitates rigorous tracking of cumulative privacy loss over multiple communication rounds. Traditional composition theorems, such as the basic and advanced composition theorems, often yield loose bounds that scale poorly with the number of iterations T , typically growing as $O(\sqrt{T})$ or $O(T)$, thereby severely restricting the feasible number of training rounds within a fixed privacy budget. To address this limitation, recent advancements have shifted toward refined accounting frameworks that leverage the properties of specific noise mechanisms and subsampling strategies to provide significantly tighter bounds.

A pivotal development in this domain is the adoption of Rényi Differential Privacy (RDP) and the moments accountant method. Unlike standard (ϵ, δ) -DP, which tracks worst-case privacy loss, RDP measures the Rényi divergence between output distributions on neighboring datasets, offering superior composition properties where privacy costs add linearly in the Rényi order α . This framework allows for the precise translation of privacy loss back to (ϵ, δ) -DP parameters only at the final stage of analysis, minimizing accumulation errors. The efficacy of this approach in deep learning contexts was notably demonstrated in foundational work on deep learning with differential privacy, which introduced the moments accountant to track the privacy cost of noisy SGD with subsampling, enabling the training of complex neural networks under modest privacy budgets [2]. Further refinements have extended these concepts to the federated setting, where [36] introduced a framework based on Gaussian differential privacy (a specific instance of f -DP) that provides analytically tractable expressions for privacy guarantees under composition and subsampling, surpassing the tightness of RDP in certain regimes. Similarly, [6] highlighted the weaknesses of traditional DP definitions in handling composition and subsampling, advocating for f -DP to derive tighter, analytically solvable bounds for optimizers like Adam and SGD without requiring sophisticated numerical integration.

Complementing these divergence-based approaches, Concentrated Differential Privacy (CDP) and its zero-concentrated variant (zCDP) have emerged as powerful tools for analyzing iterative algorithms utilizing the Gaussian mechanism. These definitions characterize privacy loss via the moment generating function of the privacy loss random variable, offering natural composition rules that are particularly well-suited for the high-dimensional, iterative updates characteristic of federated optimization. [7] utilized CDP to perform refined privacy loss analysis on data batching methods, implementing dynamic budget allocators that improve model quality under a given privacy constraint. Moreover, the integration of these accounting methods with secure aggregation protocols introduces additional complexity; [53] proposed a discrete mechanism compatible with secure multiparty computation, providing sharp bounds on Rényi divergence for shifted Skellam distributions to ensure tight privacy accounting in discrete domains.

Despite these theoretical advances, a significant gap often remains between worst-case theoretical bounds and empirical privacy leakage. Recent works suggest that theoretical guarantees can be overly conservative, prompting the development of empirical auditing techniques. [63] and [46] utilize data poisoning and adversary instantiation to establish lower bounds on privacy loss, revealing that practical leakage may be substantially lower than theoretical upper bounds suggest. Furthermore, [47] and [37] introduce methodologies for estimating privacy loss during a single training run using canary samples, providing data-dependent estimates that are often 4-5 times tighter than theoretical predictions. Looking forward, the synthesis of tight theoretical accounting via f -DP or RDP with efficient empirical auditing mechanisms represents a critical frontier. Future research must focus on bridging the divergence between conservative theoretical composition bounds and observed empirical resilience, potentially through adaptive accounting strategies that dynamically adjust privacy parameters based on real-time leakage estimates, as hinted at by [64].

3.4 Utility-Enhancing Optimizers and Adaptive Noise Strategies

Building upon the precise accounting frameworks and empirical auditing techniques established in the preceding section, the practical deployment of differentially private federated learning (DPFL) must now confront the inevitable utility gap introduced by these rigorous privacy guarantees. This degradation, primarily driven by the variance of injected noise and the bias from gradient clipping, necessitates a shift from static privacy mechanisms to dynamic, utility-enhancing strategies that adapt to the training trajectory and optimization landscape. A primary avenue of innovation involves adaptive noise scheduling and dynamic budget allocation, which recognize that the sensitivity of gradients and the tolerance for perturbation evolve throughout the training process. While static approaches often apply uniform noise levels—leading to excessive perturbation in early stages or insufficient privacy in later rounds—[65] proposes dynamically adjusting clipping thresholds and noise powers to stabilize updates, demonstrating significant accuracy improvements in strong privacy regimes. Similarly, [66] introduces an adaptive scheme that accounts for gradient heterogeneity across clients and rounds, adding decreasing noise as the model converges. Complementing this, [7] implements a dynamic privacy budget allocator to optimize the trade-off between privacy loss and model quality over time. Furthermore, [54] analytically derives a geometric series for noise amplitude, preventing premature convergence caused by persistent high-magnitude perturbations.

Beyond temporal adaptation, modifying the optimization objective itself has emerged as a potent strategy to enhance robustness against DP-induced noise. Standard optimizers often converge to sharp minima, which are highly sensitive to the parameter perturbations inherent in DP. To address this, [55] introduces DP-FedSAM, which integrates Sharpness Aware Minimization (SAM) to seek flat loss landscapes. By explicitly minimizing the sharpness of the loss function, DP-FedSAM reduces the norm of local updates and improves stability under noise, achieving state-of-the-art performance. This synergizes with findings in [67], which suggest that classic regularization techniques can sometimes offer superior privacy-preserving properties compared to standard DP-SGD. Additionally, advanced linear operators have been explored to correlate noise optimally; [68] and [69] demonstrate that injecting linearly correlated noise via matrix factorization mechanisms can significantly outperform independent noise addition, particularly in settings where privacy amplification by shuffling is infeasible.

Pre-training and transfer learning strategies offer another paradigm for utility enhancement by reducing the dependency on private data for feature extraction. [70] demonstrates that fine-tuning pre-trained foundation models with DP can achieve accuracies comparable to non-private classifiers, effectively bypassing the need to learn low-level features from noisy private gradients. However,

[71] critically notes the risks of relying on web-scraped public data, urging caution in declaring such models “private” without rigorous domain adaptation analysis. In the context of large transformer models, [72] employs parameter-efficient fine-tuning (PEFT) via dynamic low-rank adaptation, reducing the dimensionality of contributions to minimize noise impact while maintaining differential privacy guarantees. Finally, compression techniques are being re-evaluated not just for efficiency but as privacy amplifiers. [73] and [12] show that combining sparsification with gradient perturbation can amplify privacy guarantees, allowing for reduced noise magnitude while maintaining communication efficiency. Collectively, these algorithmic advancements signify a maturation of DPFL from rigid noise addition to sophisticated, context-aware optimization frameworks. While these strategies effectively mitigate the immediate utility loss, they set the stage for addressing the broader systemic interdependencies discussed in the following section, where the integration of DP intersects with critical challenges in robustness, fairness, and data heterogeneity.

3.5 Robustness, Fairness, and Heterogeneity in Private Optimization

The integration of differential privacy (DP) into federated optimization introduces complex interdependencies with system robustness, algorithmic fairness, and data heterogeneity. While DP mechanisms are primarily designed to bound information leakage, their structural properties—specifically gradient clipping and noise injection—fundamentally alter the optimization landscape, creating a trilemma between privacy, utility, and reliability. A critical insight is the dual role of DP as both a defensive shield and a potential source of vulnerability. Theoretically, DP provides certified robustness against poisoning and backdoor attacks by bounding the influence of any single participant; as demonstrated in [9], both local and central DP variants effectively mitigate backdoor infiltration, often outperforming specialized robustness defenses. Furthermore, [74] establishes that for canonical estimation tasks, there is no additional statistical cost to simultaneously guaranteeing privacy and robustness, suggesting an inherent synergy. However, this protective capacity is not absolute; [75] notes that while DP obscures malicious updates, it can also mask Sybil attacks where adversaries manipulate aggregation through fake identities, necessitating hybrid defenses that combine cryptographic verification with privacy mechanisms [76].

Concurrently, the application of uniform DP constraints in heterogeneous environments exacerbates fairness disparities. The indiscriminate addition of noise and aggressive gradient clipping disproportionately degrades model performance for underrepresented groups or clients with small, non-IID datasets. [77] highlights that measuring these disparities requires novel locally differentially private mechanisms to estimate group performance without leaking sensitive attributes. To address this, [78] proposes clustering clients based on model updates and loss values to identify sub-populations, thereby allowing for tailored privacy budgets that reduce the disparate impact of noise. Similarly, [79] resolves the conflict between fairness and privacy by integrating secure multiparty computation with DP, enabling the use of sensitive attributes for fairness constraints without exposing individual data. The challenge is further compounded in cross-silo settings, where [80] argues that standard client-level DP is ill-suited for record-level protection, advocating for personalized privacy targets that incentivize collaboration to mitigate noise-induced utility loss.

Addressing data heterogeneity requires moving beyond static noise injection. [58] introduces DP-SCAFFOLD, which incorporates control variates to correct for client drift, demonstrating superior convergence over DP-FedAvg in highly non-IID settings. Adaptive strategies are also pivotal; [66] employs adaptive gradient clipping and decreasing noise schedules aligned with convergence trends to accommodate gradient heterogeneity across clients. Moreover, [81] presents Robust-HDP, which estimates true noise levels to optimize aggregation weights, ensuring stability even when clients re-

port falsified privacy parameters. Ultimately, achieving a balance requires dynamic, context-aware mechanisms. Future research must focus on unified frameworks that jointly optimize for privacy, fairness, and robustness, potentially leveraging personalized federated learning architectures like [82] to decouple global consensus from local adaptation, thereby insulating personal models from the adverse effects of global noise injection while maintaining rigorous privacy guarantees for shared representations.

4 Hybrid Cryptographic Protocols and Secure System Architectures

4.1 Cryptographic Aggregation Primitives and Differential Privacy Synergy

The integration of Secure Multi-Party Computation (SMPC) with Differential Privacy (DP) represents a pivotal evolution in privacy-preserving federated learning, addressing the inherent limitations of deploying either mechanism in isolation. While SMPC protocols, such as secure aggregation, ensure that the server learns only the sum of client updates without accessing individual contributions, they lack formal guarantees against inference attacks derived from the aggregate itself [48]. Conversely, standard DP mechanisms introduce noise to mask individual contributions but often require a trusted aggregator to achieve optimal utility under central DP assumptions. The synergy of these primitives enables a distributed DP framework where cryptographic masking facilitates privacy amplification, significantly reducing the magnitude of noise required to satisfy rigorous privacy bounds while protecting against semi-honest and malicious servers.

A primary theoretical advancement in this domain is the realization that secure aggregation can transform local noise addition into a central DP guarantee through the shuffling or aggregation effect. By employing additive secret sharing or homomorphic encryption, clients can inject discrete Gaussian or Skellam noise locally, which, when summed securely, yields a distribution indistinguishable from central noise addition [30; 53]. This approach allows the system to achieve the utility of central DP with the trust assumptions of local DP, effectively mitigating the “noise accumulation” problem where individual noise variances would otherwise scale linearly with the number of participants. Furthermore, recent frameworks demonstrate that combining DP with SMPC reduces the growth of noise injection as the number of parties increases, balancing trade-offs between privacy, utility, and communication overhead more effectively than standalone approaches [83; 35].

However, the practical deployment of these hybrid systems faces significant challenges regarding efficiency and robustness. Traditional SMPC protocols often incur quadratic communication complexity, creating bottlenecks for large-scale networks. Innovations such as Turbo-Aggregate and FastSecAgg have reduced this overhead to $O(N \log N)$ or linear complexity by leveraging coding techniques and FFT-based secret sharing, making them viable for high-dimensional deep learning models [84; 85]. Additionally, the integration of zero-knowledge proofs allows for the verification of norm clipping and update validity without revealing underlying gradients, thereby hardening the system against poisoning attacks that could otherwise exploit the privacy noise to hide malicious contributions [86; 87]. Despite these advances, vulnerabilities persist; for instance, malicious servers may still elude secure aggregation through model inconsistency attacks if parameter validation is insufficient, highlighting the need for rigorous consistency checks alongside cryptographic protocols [62].

Looking forward, the frontier of this synergy lies in optimizing the privacy-utility-communication trilemma through advanced compression and correlated noise mechanisms. Emerging research suggests that privacy can be further amplified via compression, where randomized quantization serves as an intrinsic noise source, reducing the need for external perturbation [12]. Moreover, the

use of correlated noise in decentralized settings offers a pathway to match central DP utility without requiring a trusted aggregator, provided that pairwise secrets can be efficiently established [88; 89]. Ultimately, the convergence of cryptographic primitives and differential privacy is not merely additive but multiplicative, enabling scalable, robust, and formally private learning architectures capable of withstanding sophisticated adversarial strategies in untrusted environments.

4.2 Hardware-Enforced Confidentiality and Trusted Execution Environments

Complementing the algorithmic synergy of Secure Multi-Party Computation (SMPC) and Differential Privacy (DP) discussed previously, Trusted Execution Environments (TEEs) introduce a pivotal architectural shift by offering hardware-enforced isolation to defend against privileged software adversaries. Unlike purely cryptographic approaches that often incur prohibitive computational overheads, TEEs such as Intel SGX and ARM TrustZone create secure enclaves where code and data remain confidential and integrity-protected, even from the host operating system or cloud provider. This capability fundamentally alters the trust model in federated learning (FL), transitioning from a fully untrusted server assumption to one constrained by hardware guarantees. The [90] framework exemplifies this by offloading both local training and server-side aggregation to TEEs, shielding model updates from reconstruction and property inference attacks while incurring modest system overheads. By securing the aggregation logic within an enclave, [90] demonstrates that raw gradients need never be exposed in plaintext to the untrusted host, effectively relaxing the noise requirements for central DP and enhancing the privacy-utility trade-off.

The integration of TEEs with DP creates a robust, multi-layered privacy guarantee that addresses the limitations of standalone mechanisms. While DP provides mathematical bounds on information leakage regardless of computational power, it often requires significant noise injection that degrades model utility, particularly in high-dimensional settings. TEEs mitigate this by ensuring the aggregation process itself is opaque, allowing for the implementation of “honest-but-curious” server assumptions with verifiable integrity. This is critical because, as noted in [91], basic FL systems lack protections against malicious service providers, and relying on DP alone may necessitate excessive noise or blind trust in the provider’s correct implementation. By leveraging TEEs, systems can achieve externally verifiable privacy properties, bolstering robustness without the scalability challenges inherent in SMPC. Furthermore, the [92] framework illustrates how combining TEEs with DP primitives enables mutually distrusting parties to train deep learning models efficiently, establishing a baseline for fully private machine learning that balances security with performance.

However, the deployment of TEEs in FL faces distinct challenges regarding resource constraints and side-channel vulnerabilities, distinguishing them from the communication bottlenecks of cryptographic methods. The limited memory size of current enclaves often necessitates innovative training strategies, such as the greedy layer-wise training employed in [90], to accommodate large neural networks within the trusted area. Additionally, while TEEs protect against software attacks, they remain susceptible to side-channel analysis targeting memory access patterns or execution times. To address this, hybrid frameworks must combine hardware isolation with algorithmic defenses; for instance, integrating local DP on client devices with TEE-secured aggregation creates a defense-in-depth strategy that protects against both hardware side-channels and inference attacks on the final model. The [91] work further emphasizes the importance of open-sourcing enclave code to ensure transparency and allow for external auditing of the privacy mechanisms running inside the hardware.

Looking forward, the convergence of TEEs with advanced cryptographic protocols promises to

resolve the tension between privacy, utility, and efficiency, bridging the gap toward the decentralized architectures discussed subsequently. Emerging trends suggest a move towards verifiable computation where remote attestation allows clients to cryptographically verify the integrity of the server-side aggregation code before transmitting updates, establishing a chain of trust essential for untrusted cloud environments. Moreover, as foundation models grow in scale, the efficiency of TEE-based aggregation will be paramount. Future architectures may leverage TEEs to perform secure aggregation of encrypted gradients, reducing the communication burden while maintaining the rigorous privacy guarantees of DP. Ultimately, the symbiotic relationship between hardware-enforced confidentiality and differential privacy offers a pragmatic pathway toward trustworthy distributed intelligence, serving as a critical intermediate layer between algorithmic cryptography and the trustless, decentralized paradigms enabled by blockchain technologies.

4.3 Decentralized Ledger Technologies and Byzantine-Resilient Architectures

The integration of Decentralized Ledger Technologies (DLT) into Federated Learning (FL) architectures fundamentally addresses the single-point-of-failure vulnerability inherent in centralized server models, replacing trusted coordinators with transparent, immutable consensus mechanisms. By decentralizing the aggregation process, blockchain-enabled frameworks such as [93] and [94] transform the learning topology into a peer-to-peer network where model updates are validated and recorded via smart contracts. This architectural shift not only enhances auditability but also mitigates risks associated with malicious aggregators, as demonstrated in [95], which leverages smart contracts to automatically exclude unreliable participants. However, the immutability of blockchain introduces a critical tension with privacy preservation: recording model updates on a public ledger risks permanent exposure of sensitive gradient information. To reconcile this, recent approaches embed Differential Privacy (DP) directly into the on-chain transaction layer. For instance, [96] proposes sanitizing intermediate embeddings with Local Differential Privacy (LDP) before ledger commitment, thereby preventing ledger leakage while maintaining the transparency required for trustless verification.

Beyond privacy, decentralized architectures must exhibit robustness against Byzantine adversaries who may broadcast arbitrary or poisoned updates to disrupt convergence. The absence of a central authority necessitates distributed consensus mechanisms capable of filtering malicious contributions without compromising individual privacy. Traditional robust aggregation rules often require visibility into raw updates, creating a privacy-utility paradox. Innovative solutions like [97] resolve this by combining Lagrange coded computing with verifiable secret sharing to achieve information-theoretic privacy alongside Byzantine resilience. Similarly, [98] introduces a two-server protocol that decouples input privacy from robustness, while [99] theoretically demonstrates that robust stochastic model aggregation can be decoupled from DP noise injection, allowing for simultaneous optimization of both properties. In fully decentralized settings, [100] utilizes secure multi-party computation to compute cosine similarities for outlier detection without revealing underlying gradients, effectively defending against majority Byzantine attacks.

Despite these advances, significant challenges persist regarding the scalability and communication overhead of DLT-integrated systems. The computational cost of consensus and the latency of on-chain aggregation can impede real-time learning, particularly in resource-constrained edge environments. Furthermore, the interplay between DP noise and consensus validity remains complex; excessive noise may trigger false positives in Byzantine detection protocols. Emerging trends suggest a move toward hybrid incentive mechanisms and lightweight consensus algorithms tailored for FL, as explored in [101]. Future research must focus on optimizing the trilemma of decentral-

ization, privacy, and robustness, potentially through adaptive noise scheduling that responds to network trust levels and the development of zero-knowledge proofs to verify update correctness without revealing sensitive parameters. Ultimately, the synthesis of blockchain’s trustlessness with the mathematical rigor of differential privacy offers a promising pathway toward truly autonomous and secure distributed intelligence.

4.4 Communication-Efficient Compression under Cryptographic and Privacy Constraints

Building upon the architectural trust established by Decentralized Ledger Technologies and the robustness against Byzantine adversaries discussed previously, the integration of communication compression with cryptographic protocols and differential privacy (DP) represents a critical frontier in scaling privacy-preserving federated learning (FL). While the preceding sections addressed structural vulnerabilities and consensus mechanisms, this subsection focuses on optimizing the efficiency-privacy trade-off at the data transmission layer. Compression techniques such as quantization and sparsification are essential for mitigating bandwidth bottlenecks in distributed settings; however, their interaction with privacy mechanisms introduces complex trade-offs between utility, security, and communication efficiency. Recent advancements demonstrate that compression can be designed not merely as a utility-preserving tool but as an intrinsic component of the privacy mechanism itself.

A primary paradigm shift involves leveraging randomized quantization to simultaneously achieve communication efficiency and rigorous DP guarantees. Traditional approaches often treat compression and privacy as orthogonal steps, yet [102] establishes that randomized quantization mechanisms can generate privacy-preserving noise inherently, eliminating the need for separate Gaussian noise injection while satisfying Renyi DP constraints. This synergy is further refined in [30], which proposes discretizing data and adding discrete Gaussian noise prior to secure aggregation, theoretically matching central DP accuracy with less than 16 bits of precision per value. Similarly, [53] introduces a discrete mechanism based on Poisson variables that is closed under summation, making it uniquely compatible with secure aggregation protocols under strict communication constraints.

However, the interplay between sparsification and privacy requires careful architectural design to prevent information leakage through update patterns, a concern distinct from the gradient exposure risks in blockchain ledgers. [44] warns that naive gradient sparsification or quantization may provide a false sense of security against reconstruction attacks, necessitating formal privacy accounting rather than relying on obfuscation byproducts. To address this, [73] integrates random sparsification with gradient perturbation to amplify privacy guarantees, utilizing Renyi DP for tight end-to-end accounting. Furthermore, [103] proposes bounded local update regularization combined with sparsification to restrict update norms naturally, thereby improving the privacy-utility trade-off without sacrificing the benefits of dimensionality reduction.

When combining compression with homomorphic encryption (HE), the computational overhead of operating on compressed ciphertexts becomes a limiting factor, echoing the latency challenges observed in DLT-based systems. [104] addresses this by selectively encrypting sensitive parameters, significantly reducing computation and communication costs for large foundation models. Complementing this, [105] adapts scalar quantization and pruning to the Secure Aggregation primitive, introducing “Secure Indexing” to support extreme compression (up to 40×) without meaningful utility loss. Theoretical boundaries for these interactions are delineated in [106], which characterizes the fundamental communication cost required to achieve optimal accuracy under DP and

Secure Aggregation, demonstrating that linear schemes based on sparse random projections can achieve near-optimal bit rates.

Emerging trends point towards unified frameworks that jointly optimize these competing objectives, moving beyond the isolated solutions discussed in earlier sections. [107] presents a unified framework accommodating general compression operators and local DP, achieving superior communication complexity without compromising privacy. Additionally, [12] theoretically proves that communicating partial information via compression can amplify privacy in distributed mean estimation, requiring significantly fewer bits than uncompressed protocols. Despite these advances, challenges remain in balancing the distortion introduced by aggressive compression with the noise floor required for DP, particularly in non-IID settings where update heterogeneity is high. Future research must focus on adaptive compression strategies that dynamically adjust quantization levels and sparsity patterns based on real-time privacy budget consumption and convergence dynamics, ensuring robust performance across diverse network conditions and laying the groundwork for the next generation of efficient, privacy-centric FL architectures.

5 Navigating Heterogeneity, Fairness, Robustness, and Personalization

5.1 Personalized Learning Architectures under Differential Privacy Constraints

The integration of differential privacy (DP) into federated learning (FL) introduces a critical tension: the noise injection required for formal privacy guarantees often exacerbates the utility degradation inherent in non-IID data distributions, thereby undermining the efficacy of global model consensus. To address this, advanced personalized learning architectures have emerged, fundamentally shifting the paradigm from a single global model to frameworks that decouple shared representation learning from local adaptation. This subsection analyzes how such decoupling strategies mitigate the privacy-utility trade-off by restricting the sensitivity of shared parameters while allowing unconstrained or less-constrained local personalization.

A primary architectural strategy involves parameter decoupling, where the model is split into global feature extractors and local prediction heads. By sharing only the differentially private global representations, clients can adapt local layers to their specific data distributions without incurring additional privacy costs. [108] demonstrates that fixing the base layers and personalizing only the top layers significantly combats statistical heterogeneity. Extending this to privacy-preserving contexts, [109] proposes a representation federated learning objective where only the consensus representation is refined under DP constraints. Theoretical analysis in [109] indicates that this approach improves the utility-privacy trade-off by a factor of $\sqrt{d}$ compared to full-model sharing, as the sensitivity is confined to the lower-dimensional representation space. Similarly, [110] advances this by simultaneously learning global and personalized feature information, ensuring that collaborative benefits are retained while local specificity is preserved under privacy bounds.

Beyond static decoupling, meta-learning and initialization strategies offer dynamic pathways for personalization. Differentially private Model-Agnostic Meta-Learning (DP-MAML) aims to learn a robust global initialization that requires minimal gradient steps for local adaptation, thereby reducing the cumulative privacy budget consumption associated with multiple communication rounds. [111] provides generalization guarantees showing that coordinating such local learning with private centralized updates yields superior trade-offs. Furthermore, [72] addresses the challenge of fine-tuning large transformer models on-device by combining dynamic low-rank adaptation with DP. By reducing the dimensionality of contributions through parameter-efficient fine-tuning, this approach mitigates the performance degradation typically observed when applying DP to high-

capacity models, achieving negligible accuracy loss even under stringent ϵ budgets.

Clustering and similarity-aware aggregation constitute another vital frontier. Instead of enforcing a single global model, these methods group clients based on data similarity or task affinity to apply cluster-specific DP mechanisms. [78] highlights that clustering based on both model updates and loss values can effectively identify sub-populations, ensuring that DP noise does not disproportionately harm minority groups. [112] further refines this by aggregating models from similar clients to form personalized feature extractors, utilizing local classifiers to estimate similarity without exposing raw data. In cross-silo settings, [80] argues that mean-regularized multi-task learning serves as a strong baseline, where stronger privacy requirements incentivize silos to federate more intensely to counteract noise, thus turning privacy constraints into a catalyst for collaboration.

Despite these advancements, challenges remain in dynamically allocating privacy budgets across global and local components and in verifying personalization without leaking membership information. Future research must explore adaptive mechanisms that modulate noise levels based on local convergence rates and develop theoretical bounds for personalized DP that account for the reduced sensitivity of decoupled architectures. The synthesis of cryptographic protocols with these personalized architectures, as hinted at in [113], may further enable collaborative learning where parties improve local models without agreeing on a central architecture, marking a significant evolution toward truly trustworthy and adaptable distributed intelligence.

5.2 The Interplay Between Algorithmic Fairness and Differential Privacy

While the preceding discussion highlighted strategies to mitigate the utility degradation caused by data heterogeneity through personalized architectures, an equally critical challenge arises from the interaction between differential privacy (DP) and algorithmic fairness. Mechanisms designed to protect individual confidentiality, such as noise injection and gradient clipping, can inadvertently exacerbate disparities among demographic groups. This phenomenon occurs because minority groups, typically possessing fewer data samples, contribute gradients with higher variance; consequently, uniform clipping thresholds distort these updates more severely than those from majority groups, leading to biased global models [70]. Empirical studies confirm that private federated learning can accentuate existing unfairness, creating a “double penalty” where marginalized groups suffer both from data scarcity and the obscuring effects of privacy noise [114].

Theoretical analyses reveal that the privacy-utility trade-off is not uniform across the data distribution. In settings with heterogeneous data, the signal-to-noise ratio for minority classes diminishes rapidly under standard DP-SGD, causing classifiers to fail on these subpopulations even when overall accuracy remains acceptable [3]. This disparate impact is further compounded in vertical federated learning scenarios, where feature splitting can obscure fairness-sensitive attributes, making it difficult to audit or mitigate bias without violating privacy constraints [115]. Consequently, the assumption that privacy and fairness are orthogonal objectives is flawed; rather, they are deeply coupled, with naive privacy implementations actively undermining equitable outcomes [116].

To address these challenges, recent research has pivoted toward joint optimization frameworks that simultaneously enforce fairness constraints and differential privacy guarantees. One promising direction involves the adaptation of the method of differential multipliers to the private federated setting, allowing for the imposition of group fairness constraints while maintaining formal privacy bounds [114]. Additionally, approaches like FEDERATE combine adversarial training with differential privacy to learn text representations that are both private and fair, demonstrating that privacy and fairness can mutually reinforce each other when explicitly co-optimized [117]. Another

innovative strategy employs robust clustering to identify client subpopulations based on model updates and loss values, enabling cluster-specific privacy budgets that mitigate the disparate impact of uniform noise addition [78]. This clustering-based approach serves as a vital bridge to robustness-oriented interventions, as the identification of coherent subpopulations not only aids fairness but also lays the groundwork for defending against adversarial manipulations.

Furthermore, the measurement of fairness itself must be privatized to prevent the leakage of sensitive demographic information during audit processes. Novel locally differentially private mechanisms have been developed to estimate performance disparities across groups without exposing individual group membership, proving that accurate fairness auditing is compatible with strong privacy guarantees [77]. Similarly, specialized algorithms now allow for the private computation of evaluation metrics like AUC, ensuring that model assessment does not become a vector for label inference [118]. Looking forward, the field must move beyond static privacy budgets toward adaptive schemes that allocate privacy resources based on group representation and model convergence dynamics. The emergence of individualized privacy guarantees, where data holders consent to varying privacy levels, offers a pathway to improve utility for critical subpopulations without compromising the collective privacy floor [119]. Ultimately, achieving trustworthy distributed intelligence requires treating fairness not as an afterthought, but as a core constraint in the design of differentially private optimization algorithms, setting the stage for examining how these privacy mechanisms interact with system robustness against adversarial threats.

5.3 Robustness Against Poisoning and Adversarial Exploitation

The integration of Differential Privacy (DP) into Federated Learning (FL) introduces a complex duality regarding system robustness: while DP mechanisms inherently limit the influence of individual data points, thereby offering certified resistance against poisoning, the stochastic noise injected can simultaneously obscure malicious updates, facilitating sophisticated adversarial exploitation. Theoretically, DP provides a rigorous bound on the sensitivity of the learning algorithm to any single training sample or client contribution. This property has been formalized to derive certified robustness guarantees, where the probability of an adversarial user successfully altering the global model’s prediction is bounded by the privacy parameter ϵ [120]. Specifically, increasing the level of privacy protection (lowering ϵ) has been shown to enhance certified attack inefficacy, although this does not monotonically improve certified prediction accuracy due to the accompanying utility loss [120]. Empirical evaluations confirm that both Local and Central DP variants effectively defend against backdoor attacks, often outperforming dedicated robustness defenses that lack formal privacy guarantees [9].

However, this defensive capability is counterbalanced by vulnerabilities arising from the obfuscation nature of DP. The noise injected to satisfy (ϵ, δ) -DP can mask the statistical anomalies typically used to detect poisoned updates, enabling adversaries to launch Sybil attacks where multiple fake identities manipulate the aggregation process undetected [32]. In such scenarios, a malicious server or colluding clients can exploit the power imbalance inherent in FL protocols to reconstruct individual data points even when Distributed DP and Secure Aggregation are employed, provided the adversary controls a sufficient number of Sybil devices [32]. Furthermore, adaptive poisoning strategies have been developed that specifically target the noise distribution of DP-SGD, demonstrating that standard defensive assumptions may underestimate privacy risks when an adversary with auxiliary knowledge [121]. The interplay is further complicated in non-IID settings, where the natural heterogeneity of client data makes distinguishing between legitimate distributional shifts and adversarial perturbations significantly more challenging under DP constraints [10].

To mitigate these conflicting dynamics, recent research advocates for hybrid architectures that decouple privacy preservation from robust aggregation. Approaches such as robust stochastic model aggregation allow the system to defend against Byzantine attacks via sign-based updates while applying DP mechanisms separately to perturb these signs, ensuring that the robustness and privacy influences on learning performance remain distinct [99]. Additionally, frameworks integrating Zero-Knowledge Proofs (ZKPs) with DP enable the verification of update integrity—such as norm clipping compliance—without revealing the underlying gradients, thus preventing poisoned contributions from entering the aggregation while maintaining confidentiality [86; 122]. Other innovative protocols leverage correlated noise in decentralized settings to match central DP utility while resisting external eavesdroppers and curious peers, though controlling noise accumulation in sparse networks remains a critical challenge [88]. Ultimately, achieving optimal robustness requires navigating a fundamental trilemma where privacy, robustness, and utility are inextricably linked; theoretical lower bounds suggest that no algorithm can simultaneously minimize error from adversarial workers and DP noise without incurring a statistical cost proportional to the dimensionality of the model [10]. Future directions must therefore focus on adaptive noise scheduling and verifiable secure aggregation protocols that dynamically adjust to threat levels without compromising the formal guarantees of differential privacy.

5.4 Federated Unlearning and the Right to be Forgotten

Building upon the challenges of maintaining model integrity and trust during training, the need to dynamically remove specific data contributions has given rise to the emerging paradigm of Federated Unlearning (FU). Driven by stringent data protection regulations such as the GDPR and its mandated “Right to be Forgotten,” FU extends the privacy guarantees of Federated Learning (FL) into the post-training lifecycle. Unlike centralized machine unlearning, FU operates under unique constraints where the server lacks direct access to client data and must navigate stochastic, heterogeneous, and interactive training dynamics [123]. The primary objective is to efficiently remove the influence of a specific client or data subset from the global model while preserving the utility derived from remaining participants, a challenge that renders naive full retraining computationally prohibitive given the distributed nature of FL [124].

Current methodologies for FU can be broadly categorized into exact unlearning, which guarantees statistical indistinguishability from a model trained *ab initio*, and approximate unlearning, which seeks to bound the residual influence of deleted data. Exact approaches often leverage Total Variation (TV) stability to modify aggregation protocols, ensuring that the unlearned model converges to the same distribution as a retrained model with reduced communication overhead [125]. Conversely, approximate methods frequently employ influence functions or gradient ascent to negate specific updates. For instance, [126] proposes subtracting accumulated historical updates and utilizing knowledge distillation to restore model performance without requiring client participation, while [127] introduces a rapid retraining strategy initialized by a locally unlearned client model. More advanced techniques, such as subspace-based unlearning, eliminate target contributions by performing gradient ascent in the orthogonal space of the remaining clients’ gradients, thereby minimizing storage requirements for update histories [128]. Furthermore, feature-level unlearning has gained traction for removing sensitive attributes or backdoors by optimizing feature sensitivity, as demonstrated by the Ferrari framework [129].

However, the unlearning process itself introduces novel privacy vulnerabilities that echo the robustness concerns discussed previously. The act of deleting data can inadvertently leak information if adversaries compare pre- and post-unlearning model states to infer membership or reconstruct

deleted records, a risk termed “deletion inference.” To mitigate this, verification protocols are essential. [130] establishes a framework where leaving participants actively verify unlearning efficacy through marker injection and performance checking, granting them a “Right to Verify.” Additionally, blockchain-enabled architectures utilizing Chameleon hash functions provide immutable audit trails to prove data removal without retaining historical parameters that could be exploited [131]. The integration of Secure Aggregation (SecAgg) within clustering-based unlearning schemes further enhances privacy against dynamic user participation, ensuring that gradient leakage is prevented even during the deletion phase [132].

Despite these advances, significant challenges persist. Many existing methods rely on strong assumptions regarding data homogeneity or require the server to store extensive update histories, which conflicts with privacy-preserving goals [133]. Moreover, the trade-off between unlearning efficiency, model utility, and verifiable privacy guarantees remains an open optimization problem. Future research must focus on developing training-free unlearning mechanisms with provable skew resilience [134] and constructing unified frameworks that simultaneously address heterogeneity, robustness against poisoning, and rigorous privacy auditing in dynamic FL environments.

6 Domain-Specific Applications, Evaluation Frameworks, and Regulatory Compliance

6.1 Sector-Specific Deployments and Domain Constraints

The practical deployment of Differentially Private Federated Learning (DPFL) is profoundly shaped by the idiosyncratic constraints of high-stakes domains, where the theoretical privacy-utility trade-off is modulated by regulatory mandates, data heterogeneity, and infrastructural limitations. In healthcare, the imperative to collaborate across institutional silos without compromising patient confidentiality has driven the adoption of DPFL for tasks ranging from brain tumor segmentation to multi-site fMRI analysis. However, the sensitivity of medical imaging data necessitates rigorous protection against gradient inversion attacks, often requiring substantial noise injection that can degrade diagnostic accuracy. Studies on privacy-preserving brain tumor segmentation demonstrate that while DPFL enables collaborative training on diverse datasets like BraTS, achieving clinically viable performance requires careful calibration of the privacy budget ϵ to balance model utility against the risk of reconstructing patient records [135]. Furthermore, the non-IID nature of medical data across hospitals exacerbates convergence issues, prompting the integration of domain adaptation techniques to mitigate distributional shifts while maintaining formal privacy guarantees [136]. Emerging frameworks like OpenFL and FLOP further illustrate the shift towards partial network sharing and modular architectures to reduce communication overhead and privacy leakage in computational pathology and COVID-19 diagnosis [137; 138].

In the financial sector, the landscape is characterized by vertical data partitioning, where institutions hold complementary features for shared user bases. Here, DPFL is critical for cross-institutional fraud detection and credit scoring, yet it faces unique challenges in aligning identities and protecting proprietary feature representations. The integration of Differential Privacy with Secure Multi-Party Computation (SMPC) has emerged as a robust solution, allowing banks to collaboratively train models on encrypted gradients while obscuring individual contributions from semi-honest servers [35]. Recent advancements, such as the Starlit mechanism, address scalability and identity alignment issues, enabling efficient fraud detection across global financial networks without assuming prior account freezing or trusted third parties [139]. Nevertheless, the tension between stringent financial confidentiality standards and the utility loss induced by DP noise re-

mains a significant barrier, necessitating hybrid protocols that leverage vertical federated learning structures to minimize information leakage during feature fusion [18; 22].

Conversely, edge intelligence and IoT ecosystems present a distinct set of constraints dominated by resource scarcity and communication bandwidth limitations. In applications like Internet of Vehicles (IoV), where devices must report real-time traffic data while preserving location privacy, Local Differential Privacy (LDP) is often preferred over central models to eliminate trust in the aggregator. Innovative mechanisms, such as the Three-Outputs and hybrid quantization schemes, have been proposed to encode perturbed gradients into minimal bit representations, simultaneously achieving strong LDP guarantees and reducing transmission costs [140]. The superposition nature of wireless channels further offers a dual benefit, enabling over-the-air aggregation that inherently amplifies privacy through noise averaging while optimizing spectral efficiency [141]. However, the heterogeneity of device capabilities and data distributions in IoT networks can lead to significant fairness disparities, where uniform noise addition disproportionately affects clients with limited data or computational power [142].

Finally, the deployment of DPFL in Natural Language Processing (NLP) for conversational agents highlights the challenge of preventing memorization of sensitive user utterances in large-scale transformer models. Production systems, such as those powering next-word prediction on mobile keyboards, have successfully implemented DP-FedAvg and DP-FTRL to provide formal user-level privacy guarantees, albeit with careful attention to clipping norms and participation sampling [143; 144]. Despite these successes, recent adversarial studies reveal that transformer architectures remain vulnerable to sophisticated reconstruction attacks that exploit token embeddings, suggesting that current DP mechanisms may need to be augmented with architectural modifications like frozen embeddings or parameter-efficient fine-tuning to ensure robust privacy [145; 45]. Collectively, these sector-specific analyses underscore that effective DPFL deployment requires a nuanced synthesis of algorithmic mechanisms, cryptographic protocols, and domain-aware engineering to navigate the complex interplay of privacy, utility, and operational feasibility.

6.2 Comprehensive Evaluation Metrics and Benchmarking Frameworks

Building upon the sector-specific deployment challenges outlined previously, evaluating Differentially Private Federated Learning (DPFL) systems necessitates a departure from conventional accuracy-centric metrics toward a holistic taxonomy that rigorously quantifies the interplay between privacy guarantees, model utility, communication efficiency, and fairness. The fundamental challenge lies in characterizing the “Privacy-Utility-Communication Trilemma,” where enhancing one dimension often incurs substantial costs in others. Recent theoretical work establishes that there is effectively “no free lunch” in this triad; quantitative bounds demonstrate that minimizing privacy leakage via mechanisms like randomization or homomorphic encryption invariably degrades utility or increases communication overhead [5; 4]. Consequently, robust benchmarking frameworks must simultaneously report the privacy budget (ϵ, δ), convergence rates, and transmission costs to provide a complete system assessment that reflects the operational realities discussed in healthcare, finance, and IoT domains.

A critical component of modern evaluation is the empirical auditing of privacy claims, as theoretical upper bounds on privacy loss are often conservative and may not reflect practical leakage. Studies reveal a significant gap between worst-case theoretical guarantees and the effective privacy loss measured through adversarial instantiation, suggesting that standard implementations frequently operate with vacuous privacy guarantees at useful utility levels [3; 46]. To address this discrepancy

and bridge the gap between formal guarantees and actual security, advanced methodologies such as CANIFE utilize carefully crafted “canary” samples to empirically measure per-round exposure, yielding tight, data-dependent privacy estimates that are significantly lower than theoretical bounds [37]. Furthermore, one-shot empirical estimation techniques now allow for efficient privacy auditing during a single training run, overcoming the computational prohibitions of retraining-based audits [47]. These empirical metrics are essential for validating that mechanisms like the Skellam mechanism or banded matrix factorization deliver promised privacy-accuracy trade-offs in realistic deployments [53; 146].

Beyond privacy and utility, benchmarking must account for fairness and heterogeneity, particularly in non-IID settings where uniform noise injection can disproportionately degrade performance for underrepresented groups—a concern echoed in the disparities observed in edge intelligence and medical data distributions. Research indicates that DPFL introduces disparate impacts, necessitating metrics that evaluate demographic parity and equalized odds alongside global accuracy [147]. Innovative approaches, such as robust clustering based on model updates and loss values, are being developed to mitigate these disparities, requiring evaluation frameworks that can assess cluster-specific utility under varying privacy budgets [78]. Additionally, communication efficiency remains a bottleneck, especially when secure aggregation is employed. Theoretical analyses define the fundamental bit-cost per client required to achieve optimal accuracy under central DP, providing a baseline for evaluating compression-friendly protocols [106; 12].

Standardized simulation environments are pivotal for reproducible research in this domain, enabling the systematic testing of these multidimensional trade-offs. Frameworks such as APPFL, OpenFL, FedLab, and pfl-research offer modular architectures to benchmark diverse DP algorithms across heterogeneous hardware and network conditions [148; 137; 149; 150]. These platforms facilitate the comparison of adaptive noise strategies, such as amplitude-varying perturbations, against static baselines, enabling researchers to optimize the balance between premature convergence and privacy preservation [54]. Future evaluation paradigms must further integrate robustness metrics against poisoning attacks, as DP mechanisms inherently interact with Byzantine resilience, sometimes offering certified robustness while potentially masking malicious updates [9; 99]. By synthesizing these multidimensional metrics, the community can move toward trustworthy DPFL systems that are rigorously validated across all operational constraints, setting the stage for the detailed empirical auditing methodologies explored in the subsequent section.

6.3 Empirical Privacy Auditing and Attack Simulation Methodologies

Theoretical differential privacy (DP) guarantees often provide worst-case upper bounds that may be overly conservative or, conversely, vacuous in complex deep learning architectures due to implementation nuances. Consequently, empirical privacy auditing has emerged as a critical counterpart to analytical accounting, utilizing adversarial simulations to validate stated privacy bounds and measure actual information leakage. This subsection synthesizes methodologies for stress-testing Federated Learning (FL) systems, bridging the gap between formal ϵ -guarantees and practical security.

A primary approach involves instantiating hypothetical adversaries to establish lower bounds on privacy loss. [46] demonstrates that by constructing specific distinguishing games, one can empirically verify if the theoretical upper bounds of DP-SGD are tight. Their findings suggest that while theoretical bounds are often met, realistic restrictions on adversary capabilities can reveal a significant gap, indicating that standard DP analysis may be conservative. Complementing this,

[151] introduces a unified methodology combining privacy search with influence-based poisoning attacks. This framework achieves superior statistical power in detecting privacy violations across diverse models, effectively identifying implementation errors such as incorrect sensitivity clipping or flawed noise generation that nullify formal guarantees.

Beyond generic auditing, specialized attack simulations provide granular insights into leakage mechanisms. Membership Inference Attacks (MIAs) serve as a proxy for privacy risk, where the success rate of an adversary inferring training membership correlates with privacy loss. [152] proposes Differential Training Privacy (DTP) as an empirical metric derived from MIA success, arguing that reducing DTP directly mitigates privacy risk. In the FL context, [153] highlights that multi-temporal and multi-spatial analysis of model updates significantly enhances MIA effectiveness, necessitating robust defenses like data replacement over simple gradient perturbation. Furthermore, [154] reveals that dishonest servers can craft malicious parameters to bypass Local DP protections, achieving high inference success rates that theoretical bounds fail to predict under active threat models.

To obtain tighter, data-dependent privacy estimates, canary-based insertion strategies have gained prominence. [37] introduces a novel method where carefully crafted trap samples are inserted into training sets to quantify empirical exposure per communication round. Their results indicate that empirical per-round ϵ can be 4–5 times lower than theoretical worst-case bounds in realistic FL scenarios, suggesting that theoretical accounts often overestimate leakage. Similarly, [49] utilizes Generative Adversarial Networks (GANs) to reconstruct data from degraded gradients, demonstrating that even with defensive noise, latent space priors can facilitate high-fidelity reconstruction, thereby challenging the efficacy of standard gradient compression defenses.

Information-theoretic approaches offer an alternative perspective by modeling the learning process as a communication channel. [155] establishes bounds on reconstruction errors based on the mutual information between local datasets and transmitted parameters, providing a rigorous foundation for constraining leakage. Additionally, [156] derives direct bounds for membership and attribute inference without relying on the indirection of (ϵ, δ) -DP, offering faster and more interpretable privacy evaluations.

Despite these advancements, challenges remain in scaling auditing to large language models and heterogeneous cross-silo settings. [157] exposes substantial vulnerabilities in adapted FL protocols for LLMs, where active inference attacks succeed even under state-of-the-art DP mechanisms. Future research must focus on developing efficient, one-shot auditing techniques like those proposed in [47], which estimate privacy loss during a single training run, to make empirical validation feasible for resource-constrained distributed environments. Ultimately, integrating these empirical methodologies with formal verification is essential for establishing trustworthy, compliant FL systems.

6.4 Regulatory Alignment, Legal Compliance, and Ethical Governance

Building upon the empirical auditing methodologies and one-shot estimation techniques discussed previously, this subsection examines the translation of these technical validations into legally defensible privacy engineering. The integration of Federated Learning (FL) and Differential Privacy (DP) represents a pivotal shift from mere technical obfuscation to a framework capable of addressing the escalating demands of global data protection regimes. The General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) mandate rigorous anonymization and data minimization, principles that FL inherently supports by decentralizing data custody [158]. However, as empirical audits have revealed, raw parameter exchange in FL remains vulnerable to reconstruction attacks, necessitating the formal guarantees of DP to satisfy the legal thresh-

old for pseudonymization and prevent re-identification [1]. By quantifying privacy loss through the parameter ϵ , DP provides the mathematical rigor required to demonstrate compliance with the “state-of-the-art” security measures mandated by Article 32 of the GDPR, moving beyond heuristic defenses to provable anonymity [64].

Emerging regulatory frameworks, such as the EU AI Act, further complicate this landscape by imposing strict risk management and transparency obligations on high-risk AI systems. In this context, the synergy between FL and DP serves as a critical mitigation strategy, reducing the systemic risk of data breaches while maintaining model utility [38]. Nevertheless, a significant challenge lies in the “Right to be Forgotten” (Article 17 GDPR), which demands the complete erasure of user data upon request. In distributed settings, simply removing a client does not expunge their influence from the global model weights, a limitation that empirical auditing must also account for. Consequently, Federated Unlearning (FU) has emerged as an essential compliance mechanism, employing techniques such as rapid retraining, knowledge distillation, and influence subtraction to mathematically guarantee the removal of specific contributions without full model re-initialization [127; 124]. Recent advancements propose verifiable unlearning frameworks, leveraging blockchain and cryptographic proofs to provide audit trails that satisfy regulatory demands for accountability and demonstrable compliance [130; 131].

Liability distribution remains a contentious legal frontier in these decentralized architectures. Unlike centralized systems where the data controller bears sole responsibility, FL disperses computational agency across multiple silos, creating ambiguity in assigning liability for privacy leaks or model poisoning. The “No Free Lunch” theorem for privacy-utility trade-offs suggests that perfect security is unattainable, implying that legal frameworks must evolve to define acceptable risk thresholds rather than absolute immunity [5; 4]. To address this, ethical governance models are increasingly advocating for “privacy by design” workflows that integrate the empirical auditing and standardized benchmarking strategies detailed in the preceding section into the development lifecycle [151]. Future regulatory alignment will likely depend on the standardization of these auditing protocols, ensuring that technical privacy claims are not only theoretically sound but empirically verifiable against sophisticated adversarial strategies, thereby bridging the gap between algorithmic innovation and legal accountability [28].

7 Conclusion

The synergistic integration of Federated Learning (FL) and Differential Privacy (DP) has established a rigorous paradigm for distributed intelligence, fundamentally shifting the trust model from centralized data custodians to mathematically verifiable guarantees. This survey has elucidated that while FL mitigates raw data exposure, it remains vulnerable to sophisticated inference and reconstruction attacks unless fortified by the formal privacy bounds of DP [159]. The core achievement of current research lies in reconciling the inherent tension between privacy, utility, and communication efficiency, often framed as a trilemma where enhancing one dimension inevitably compromises the others [4; 5]. State-of-the-art mechanisms, such as adaptive clipping strategies like AdaClip and advanced accounting methods utilizing Rényi DP, have significantly narrowed the utility gap, enabling the training of deep neural networks with non-convex objectives under modest privacy budgets [51; 2]. Furthermore, the emergence of shuffled models and secure aggregation protocols has demonstrated that privacy amplification via subsampling can yield central-level guarantees from local randomizers, effectively reducing the noise magnitude required for convergence [53; 106].

Despite these advancements, critical limitations persist, particularly regarding the disparate impact

of uniform noise injection on model fairness and the scalability of these mechanisms to foundation models. Empirical audits reveal a substantial gap between theoretical upper bounds on privacy loss and the effective privacy measured against realistic adversaries, suggesting that current implementations often operate with overly conservative budgets that degrade utility without commensurate security gains [3; 46]. Moreover, the interplay between DP and algorithmic fairness remains contentious; noise addition disproportionately degrades performance for underrepresented groups, necessitating novel fairness-aware clipping and personalized privacy budgeting strategies to prevent the exacerbation of societal biases [114; 147]. The scalability challenge is equally pressing, as traditional DP-SGD struggles with the computational overhead of high-dimensional foundation models, prompting a shift towards matrix factorization mechanisms and banded operators that optimize the privacy-utility-computation trade-off for adaptive streams [160; 146].

Future research must pivot towards dynamic, context-aware privacy frameworks that transcend static ϵ -budgets. The development of Federated Foundation Models (FFMs) demands mechanisms capable of handling continual learning and prompt tuning while preserving privacy across heterogeneous domains [161]. Additionally, the integration of cryptographic primitives with DP, such as homomorphic encryption-enhanced noise calibration, offers a pathway to robust defense against both passive inference and active poisoning attacks, although the computational cost remains a barrier for edge deployment [104; 162]. Emerging frontiers also include the exploration of quantum-resistant DP mechanisms and the formalization of individualized privacy guarantees that accommodate varying user risk profiles, moving beyond the “one-size-fits-all” approach [119; 163]. Ultimately, achieving trustworthy distributed intelligence requires a holistic optimization of privacy, robustness, and fairness, supported by standardized empirical auditing frameworks that validate theoretical claims against evolving adversarial capabilities [151; 10].

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